



The Open University of Israel

Department of Mathematics and Computer Science

Thesis Proposal

Route-Aware ETA Prediction with Dynamic Graph Neural Networks

Thesis proposal is submitted as a partial fulfillment of the requirements
towards an M.Sc. degree in Computer Science

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1 Work Objective

This thesis aims to improve the accuracy of Estimated Time of Arrival (ETA) prediction by developing a novel dynamic Graph Neural Network (GNN) architecture that models traffic at the vehicle level with explicit route information. Unlike existing approaches that either use static sensor networks (e.g., DCRNN [1], ST-GCN [2]) or infer routes implicitly from origin-destination pairs (e.g., DeepTTE [3], STAD [4]), this work proposes a hybrid graph structure where static infrastructure (junctions as nodes, roads as edges) is combined with dynamic vehicle nodes and time-varying edges. The dynamic edges create junction-vehicle-vehicle-junction relations by connecting vehicles to junctions (traversal edges) and vehicles to other vehicles on the same road segment (interaction edges), enabling rich modeling of vehicle interactions and traffic flow dynamics.

The primary research objective is to investigate whether incorporating predefined vehicle routes as explicit route information significantly enhances ETA prediction accuracy compared to models that infer routes implicitly (e.g., DeepTTE [3], STAD [4]). Current state-of-the-art methods such as DuETA [5] use route information but focus on aggregated road segments rather than individual vehicle trajectories. This work will develop and evaluate a dynamic GNN architecture that explicitly encodes route intent at the vehicle level, comparing its performance against baseline models that infer routes implicitly (e.g., DeepTTE [3], STAD [4]) or use static sensor networks (e.g., DCRNN [1], ST-GCN [2]). The research will demonstrate the contribution of route-aware modeling combined with vehicle-level dynamic graph representation to achieving more accurate ETA predictions in dynamic traffic scenarios.

2 Background with Reference to Sources

2.1 Previous Work in Route-Aware Traffic Prediction

This research builds upon previous work in traffic prediction and route-aware modeling. Voloch and Voloch-Bloch [6] demonstrated real-time future traffic estimations for navigation route optimization, establishing the foundation for dynamic traffic prediction. Their approach used simulation-based methods to estimate future traffic conditions and optimize navigation routes accordingly. Building on this, Voloch et al. [7] introduced smart simulative route predictions that anticipate future congestion through heuristic simulation, improving ETA accuracy. These works laid the methodological groundwork for graph-based analysis of traffic dynamics and route-aware modeling, demonstrating the value of incorporating route information and future traffic predictions in navigation systems.

2.2 State-of-the-Art ETA Prediction Methods

With the growth of urban mobility datasets, machine learning models have been applied to ETA prediction. Tree-based methods such as XGBoost [8] leverage handcrafted features (trip distance, time-of-day) and perform well on aggregated trip records, including NYC Taxi [9] and Porto Taxi [10] datasets. However, they cannot capture fine-grained spatio-temporal interactions.

Trajectory- and trip-based neural models address this gap. DeepTTE [3] learns ETA from raw GPS traces and reports results on city-scale taxi/ride-hailing datasets with per-trip horizons typically within tens of minutes. STAD [4] adjusts routing-engine outputs with spatio-temporal corrections, improving travel-time estimation. Production systems such as STANN [11] evaluate on large ride-hailing ETA datasets with short-to-medium trip durations.

In parallel, graph-based spatio-temporal learning has advanced traffic prediction by representing road networks as graphs. DCRNN [1] combined diffusion graph convolutions with recurrent units and became a common backbone for traffic speed/flow prediction on sensor benchmarks (METR-LA [12], PEMS-BAY [13]). ST-GCN [2] introduced spatio-temporal graph convolutional networks for traffic forecasting, while Graph WaveNet [14] advanced deep spatial-temporal graph modeling. Notably, DCRNN-style encoders have also been adapted for ETA prediction in production settings.

Most recently, DuETA [5] introduced duration-aware ETA modeling at Baidu Maps, categorizing trips into short (0–3 km), medium (3–10 km), and long (>10 km) segments. DuETA achieved MAEs of 27s, 46s, and 98s respectively across these categories, significantly improving upon prior baselines. DuETA uses route information but focuses on aggregated road segments rather than individual vehicle trajectories with explicit route intent.

2.3 Limitations of Existing Approaches

However, existing benchmarks such as NYC [9], Porto [10], Chengdu/DiDi [15], and Geolife [16] lack explicit representation of pre-planned routes. Models must therefore infer likely paths between origin and destination, introducing ambiguity and reducing accuracy. Most ETA prediction methods either use static sensor networks (e.g., DCRNN [1], ST-GCN [2]) that aggregate traffic at fixed locations, or rely on trajectory-based learning (e.g., DeepTTE [3], STAD [4]) that infers routes implicitly from origin-destination pairs. Current state-of-the-art methods such as DuETA [5] use route information but focus on aggregated road segments rather than individual vehicle trajectories with explicit route intent.

Beyond methodological limitations, existing datasets present significant constraints that hinder comprehensive model comparison and evaluation. Public datasets such as NYC Taxi [9] and Porto Taxi [10] provide trip-level records with origin-destination pairs and GPS trajectories, but lack explicit route information and graph structures. These datasets require models to infer routes implicitly from origin-destination pairs, introducing ambiguity and reducing accuracy. Moreover, proprietary datasets used by production systems, such as the Baidu Maps dataset

employed by DuETA [5], are not publicly accessible, preventing independent validation and fair comparison across different model architectures.

The absence of a unified dataset that supports all model types creates a fundamental challenge for comprehensive evaluation. Trajectory-based models (DeepTTE [3], STAD [4]) require GPS trajectory sequences, static graph models (DCRNN [1], ST-GCN [2]) require aggregated sensor network data, and route-aware models (DuETA [5]) require route segment information. No existing public dataset provides the combination of explicit routes, dynamic graph structure, and vehicle-level data required for evaluating all model types simultaneously. This limitation necessitates the creation of a simulated dataset with multiple variants, where each variant is tailored to a specific model’s requirements while maintaining the same underlying traffic simulation, enabling fair comparison across all models.

These limitations highlight the need for approaches that: (1) model traffic at the vehicle level rather than aggregated sensor networks, (2) explicitly encode predefined routes rather than inferring them implicitly, (3) combine dynamic graph structures with temporal learning to capture evolving traffic conditions, and (4) provide a unified evaluation framework that enables fair comparison across different model architectures through a multi-variant dataset generated from the same underlying traffic simulation.

3 Work Description

This thesis will extend the DSTR-GNN architecture [17] to further investigate the contribution of explicit route information to ETA prediction accuracy and explore additional improvements to the model. The work will be conducted through systematic experimental evaluation, architectural enhancements, and comprehensive comparison with state-of-the-art baselines.

3.1 Research Methodology

The research will follow an iterative experimental methodology combining model development, ablation studies, and comparative evaluation. The work will leverage a four-week urban simulation dataset generated using SUMO [18], which provides dynamic traffic snapshots with explicit route information for each vehicle. The dataset will be partitioned chronologically to ensure temporal separation between training, validation, and test sets, capturing diverse traffic patterns including rush hours, weekends, and varying trip durations.

3.2 Multi-Variant Dataset Generation

To enable fair comparison across different model architectures, this work will generate a multi-variant dataset from a single SUMO simulation. Rather than converting a single dataset format to match different model requirements (which could introduce bias or information loss), multiple

variants will be extracted from the same underlying traffic simulation, ensuring all models are evaluated on identical traffic patterns while receiving data in their native format.

The multi-variant dataset will include four primary variants: (1) **dynamic graph variant** for DSTRA-GNN, containing dynamic graph snapshots with static junction nodes, dynamic vehicle nodes, and time-varying edges creating junction-vehicle-vehicle-junction relations; (2) **trajectory variant** for trajectory-based models (DeepTTE [3], STAD [4]), containing GPS trajectory sequences extracted from vehicle movements; (3) **static graph variant** for static sensor network models (DCRNN [1], ST-GCN [2]), containing aggregated sensor readings at fixed locations with static road network topology; and (4) **route segment variant** for route-aware models (DuETA [5]), containing trip-level records with route sequences and aggregated segment features.

This approach ensures that all models are evaluated on the same underlying traffic simulation, providing a fair comparison while respecting each model’s specific data requirements. The consistent train/validation/test splits across all variants further guarantee that performance differences reflect model capabilities rather than dataset variations.

3.3 Main Components and Stages

Stage 1: Baseline Comparison and Validation The first stage will establish comprehensive baselines by implementing and evaluating existing state-of-the-art methods using their respective dataset variants. This includes: (1) trajectory-based models that infer routes implicitly (DeepTTE [3], STAD [4]) evaluated on the trajectory variant, (2) static sensor network approaches (DCRNN [1], ST-GCN [2]) evaluated on the static graph variant, and (3) route-aware methods (DuETA [5]) evaluated on the route segment variant. The evaluation will use standard metrics (MAE, RMSE, MAPE) to establish performance benchmarks and validate the dataset’s suitability for route-aware ETA prediction research.

Stage 2: Route Information Contribution Analysis Building on the DSTRA-GNN foundation, this stage will conduct systematic ablation studies to quantify the contribution of explicit route information. Experiments will compare: (1) route-aware variants that explicitly encode predefined vehicle routes, (2) route-agnostic variants that infer routes implicitly from origin-destination pairs, and (3) hybrid approaches that combine both strategies. The analysis will measure performance differences across trip categories (short, medium, long distances) and traffic conditions (congested vs. free-flow) to identify scenarios where route information provides the greatest benefit.

Stage 3: Architectural Enhancements This stage will explore improvements to the DSTRA-GNN architecture, focusing on: (1) enhanced route encoding mechanisms beyond simple edge-ID embeddings (e.g., attention-based route sequence modeling, hierarchical route representations), (2) improved temporal aggregation strategies for capturing long-term traffic patterns, (3) refinement of the Mixture-of-Experts routing mechanism for better specialization across heterogeneous

traffic regimes, and (4) integration of additional contextual features (weather, events, historical patterns) that may influence ETA accuracy.

Stage 4: Comprehensive Evaluation and Analysis The final stage will conduct extensive evaluation of the enhanced model across multiple dimensions: (1) performance comparison with all baseline methods on standard metrics, (2) analysis of prediction accuracy across different trip characteristics (distance, duration, route complexity), (3) evaluation of computational efficiency and scalability, (4) error analysis to identify failure modes and limitations, and (5) visualization of learned representations to provide interpretability insights. The results will be analyzed to draw conclusions about the value of explicit route information and the effectiveness of the proposed enhancements.

3.4 Expected Deliverables

The work will produce: (1) an enhanced dynamic GNN architecture for route-aware ETA prediction, (2) a multi-variant dataset enabling fair comparison across different model architectures, (3) comprehensive experimental results demonstrating the contribution of route information and architectural improvements, (4) detailed ablation studies quantifying the impact of different model components, (5) comparison with state-of-the-art baselines across multiple evaluation metrics, and (6) analysis and insights regarding when and why explicit route information improves ETA prediction accuracy.

4 Work Importance

This thesis extends the foundation established in [6, 7] by developing a dynamic Graph Neural Network architecture that models traffic at the vehicle level with explicit route information. The proposed approach represents a significant advancement by combining: (1) vehicle-level dynamic graph representation with static infrastructure nodes and dynamic vehicle nodes, where time-varying edges create junction-vehicle-vehicle-junction relations that capture vehicle interactions and traffic flow dynamics, (2) explicit encoding of predefined vehicle routes, and (3) temporal learning from stable road edges across recent history.

Initial results from the DSTRA-GNN model (detailed in Appendix A [17]) demonstrate the value of this approach. On a four-week urban simulation dataset, the route-aware temporal model achieved 46.2 s MAE, representing an 82.3% improvement over the average baseline (260.6 s MAE). These results validate that explicit route intent combined with vehicle-level dynamic graph modeling significantly improves ETA prediction accuracy compared to methods that infer routes implicitly (e.g., DeepTTE [3], STAD [4]).

The theoretical contribution of this work lies in demonstrating that explicit route information is a decisive feature for ETA prediction, with route-aware models achieving over 40% lower error compared to models that infer routes implicitly (e.g., DeepTTE [3], STAD [4]) or use

static sensor networks (e.g., DCRNN [1], ST-GCN [2]). The practical value includes improved accuracy for navigation systems, fleet management, and logistics operations, enabling better travel decisions and congestion management.

This research addresses critical gaps in existing ETA prediction methods by providing a unified framework that integrates explicit route intent, dynamic vehicle-junction-vehicle graphs (with vehicle-to-vehicle interaction edges), and temporal learning, advancing the state-of-the-art in traffic prediction research.

5 Methods for Work Execution

5.1 Types of Activities

The work will involve several types of activities:

Literature Review and Analysis: Comprehensive review of existing ETA prediction methods, graph neural network architectures, and route-aware modeling approaches. This includes analyzing published papers, understanding implementation details, and identifying gaps in current research.

Software Development and Programming: Implementation of baseline models (DeepTTE [3], STAD [4], DCRNN [1], ST-GCN [2], DuETA [5]) and extension of the DSTR-GNN architecture. Development will include model architecture design, training pipelines, evaluation frameworks, and visualization tools.

Experimental Design and Execution: Systematic experimental evaluation including baseline comparisons, ablation studies, and architectural enhancements. Experiments will be conducted with proper hyperparameter tuning, cross-validation, and statistical significance testing.

Data Analysis and Interpretation: Statistical analysis of experimental results, error analysis, performance comparison across different scenarios, and visualization of learned representations. This includes identifying patterns, failure modes, and drawing insights about when route information is most beneficial.

Documentation and Writing: Documentation of implementations, experimental protocols, results analysis, and thesis writing.

5.2 Tools and Software

Programming Languages and Frameworks:

- **Python 3.8+:** Primary programming language for all implementations
- **PyTorch:** Deep learning framework for implementing and training neural network models
- **PyTorch Geometric:** Library for graph neural network operations and dynamic graph processing

- **NumPy, Pandas:** Data manipulation and numerical computations
- **Matplotlib, Seaborn:** Visualization and plotting of results

Traffic Simulation and Data:

- **SUMO (Simulation of Urban MObility) [18]:** Traffic simulation platform for generating dynamic traffic datasets with explicit route information
- **Existing Simulation Dataset:** Four-week urban simulation dataset with dynamic traffic snapshots at 30-second intervals, including vehicle trajectories, route information, and traffic state features

Development and Experimentation Tools:

- **Git:** Version control for code management
- **Jupyter Notebooks:** Interactive development and analysis
- **Weights & Biases (wandb) or TensorBoard:** Experiment tracking and visualization
- **Slurm or local GPU clusters:** Distributed training and experiment execution

Analysis and Evaluation:

- **scikit-learn:** Additional machine learning utilities and metrics
- **Statistical analysis libraries:** For significance testing and confidence intervals
- **LaTeX:** Thesis and paper writing

5.3 Data Collection and Preprocessing

The work will leverage the existing four-week SUMO simulation dataset, which includes:

- Dynamic graph snapshots at 30-second intervals with static junction nodes, dynamic vehicle nodes, static road edges (junction-to-junction), and time-varying dynamic edges. The dynamic edges include: (1) traversal edges connecting vehicles to junctions (junction-vehicle and vehicle-junction), and (2) interaction edges connecting vehicles to other vehicles on the same road segment, creating junction-vehicle-vehicle-junction relations that capture vehicle interactions and traffic flow dynamics
- Per-vehicle route information (`vehicle_route_left`) encoding the remaining path sequence
- Node features (28 dimensions): junction/vehicle type, kinematics, temporal encodings, zone information, coordinates, current-edge attributes, demand/occupancy surrogates

- Edge features (7 dimensions): average speed, lanes, length, demand, occupancy

Data preprocessing will include: (1) chronological partitioning into training (2 weeks), validation (1 week), and test (1 week) sets, (2) feature normalization and standardization, (3) temporal window construction ($H = 30$ snapshots), (4) route sequence encoding and vocabulary construction, and (5) target normalization (log-transform and z-score normalization).

5.4 Analysis Techniques

Performance Evaluation: Standard regression metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Mean Absolute Percentage Error (MAPE). Metrics will be computed per-vehicle and aggregated across different trip categories (short, medium, long distances) and traffic conditions.

Ablation Studies: Systematic component removal and replacement to quantify individual contributions: (1) route-aware vs. route-agnostic variants, (2) temporal vs. non-temporal variants, (3) dynamic vs. static graph structures, (4) different route encoding mechanisms, and (5) MoE vs. single-head prediction.

Statistical Analysis: Confidence intervals, significance testing (t-tests, Wilcoxon signed-rank tests), and effect size calculations to ensure robust conclusions about performance differences.

Error Analysis: Identification of failure modes through analysis of prediction errors across different scenarios: trip characteristics (distance, duration, route complexity), traffic conditions (congested vs. free-flow), time-of-day patterns, and spatial zones.

Interpretability Analysis: Visualization of learned representations, attention weights, expert routing patterns, and route encoding embeddings to understand model behavior and decision-making.

5.5 Validation Methods

Experimental Validation: All experiments will follow rigorous validation protocols:

- **Temporal Split:** Chronological data partitioning to prevent data leakage and ensure realistic evaluation on future time periods
- **Hyperparameter Tuning:** Grid search or Bayesian optimization on validation set with early stopping to prevent overfitting
- **Multiple Runs:** Experiments repeated with different random seeds (e.g., seeds 42, 43, 44) to ensure reproducibility and account for variance
- **Model Selection:** Selection based on lowest validation MAE, with final evaluation on held-out test set

Baseline Comparison: Comprehensive comparison with state-of-the-art methods ensures fair evaluation and validates the contribution of the proposed approach.

Reproducibility: All code, configurations, and experimental protocols will be documented and made available to ensure reproducibility of results.

6 Work Execution Timeline

The work will be executed over six months, organized into four main stages aligned with the work description. The timeline accounts for literature review, implementation, experimentation, analysis, and thesis writing.

- **Month 1: Stage 1 - Baseline Comparison and Validation**
 - Literature review and analysis of baseline methods (DeepTTE [3], STAD [4], DCRNN [1], ST-GCN [2], DuETA [5])
 - Dataset preparation and preprocessing: chronological partitioning, feature normalization, temporal window construction
 - Implementation of baseline models and evaluation framework
 - Initial baseline experiments and performance benchmarking
- **Month 2: Stage 1 Completion and Stage 2 Initiation**
 - Completion of baseline comparison experiments
 - Analysis of baseline results and validation of dataset suitability
 - Begin Stage 2: Implementation of route-aware and route-agnostic DSTRA-GNN variants
 - Design and implementation of ablation study framework
- **Month 3: Stage 2 - Route Information Contribution Analysis**
 - Systematic ablation studies comparing route-aware vs. route-agnostic variants
 - Experiments across different trip categories and traffic conditions
 - Analysis of route information contribution across scenarios
 - Statistical analysis and significance testing of results
- **Month 4: Stage 3 - Architectural Enhancements**
 - Design and implementation of enhanced route encoding mechanisms
 - Development of improved temporal aggregation strategies
 - Refinement of Mixture-of-Experts routing mechanism

- Integration of additional contextual features (if applicable)
- Training and evaluation of enhanced architecture variants
- **Month 5: Stage 4 - Comprehensive Evaluation and Analysis**
 - Final model training and hyperparameter tuning
 - Comprehensive evaluation: performance comparison, error analysis, computational efficiency assessment
 - Visualization of learned representations and interpretability analysis
 - Statistical analysis and synthesis of all experimental results
 - Identification of failure modes and limitations
- **Month 6: Thesis Writing and Finalization**
 - Writing thesis chapters: introduction, methodology, experiments, results, discussion, conclusion
 - Preparation of figures, tables, and visualizations
 - Final review and refinement of experimental results
 - Thesis editing, proofreading, and formatting
 - Final submission preparation

Note: The timeline allows for iterative refinement, where insights from later stages may inform adjustments to earlier components. Regular progress reviews with the supervisor will ensure alignment with research objectives and timely completion.

7 Initial Source List

References

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A Previous Work: Dynamic Route-Aware Graph Neural Networks for Accurate ETA Prediction

This appendix presents the paper "Dynamic Route-Aware Graph Neural Networks for Accurate ETA Prediction" by Tordjman and Voloch [17], which serves as the foundation for this thesis research. The paper introduces a dynamic spatio-temporal Graph Neural Network (GNN) architecture that explicitly encodes vehicle route intent and models traffic networks as hybrid graph structures combining static infrastructure (junctions and roads) with dynamic vehicle nodes. The graph structure includes time-varying edges that create junction-vehicle-vehicle-junction relations: traversal edges connect vehicles to junctions, and interaction edges connect vehicles to other vehicles on the same road segment, enabling rich modeling of vehicle interactions and traffic flow dynamics.

A.1 Key Contributions

The paper demonstrates that:

- Explicit route information significantly improves ETA prediction accuracy, with route-aware models achieving over 40% lower error compared to models that infer routes implicitly (e.g., DeepTTE [3], STAD [4])
- Vehicle-level dynamic graph representation captures traffic dynamics more effectively than static sensor networks (e.g., DCRNN [1], ST-GCN [2])
- Temporal learning from stable road edges provides additional refinement, achieving 82.3% improvement over baseline methods
- The combination of explicit route intent, dynamic graph structure, and temporal aggregation yields state-of-the-art results

A.2 Model Architecture

The DSTRA-GNN (Dynamic Spatio-Temporal Route-Aware GNN) architecture integrates:

- **Graph Encoder:** Multi-layer GATv2-based encoder processing dynamic graph snapshots
- **Temporal Aggregator:** Transformer-based module learning temporal context from stable road edges
- **Route Encoder:** Edge-ID embedding encoding the remaining route sequence for each vehicle
- **MoE Head:** Sparse Mixture-of-Experts for specialized prediction across heterogeneous traffic regimes

A.3 Experimental Results

On a four-week urban simulation dataset:

- Average baseline: 260.6 s MAE
- Route-aware non-temporal variant: 58.4 s MAE
- Temporal route-aware model: 46.2 s MAE (82.3% improvement)

These results validate the value of explicit route intent and vehicle-level dynamic graph modeling for ETA prediction. This thesis extends this work by further investigating the contribution of route information and exploring additional improvements to the architecture.